

UNDERSTANDING THE HETEROGENITY OF PARKINSONS DISEASE

Novel Computational and Machine Learning
Approaches Shed Light on Parkinson's Disease
Heterogeneity and Subtypes

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Abstract

Key Words: neurodegenerative, heterogeneity, PD subtypes, predictive analysis, extreme phenotypic analysis, machine learning, personalized clinical trials, multi-omics

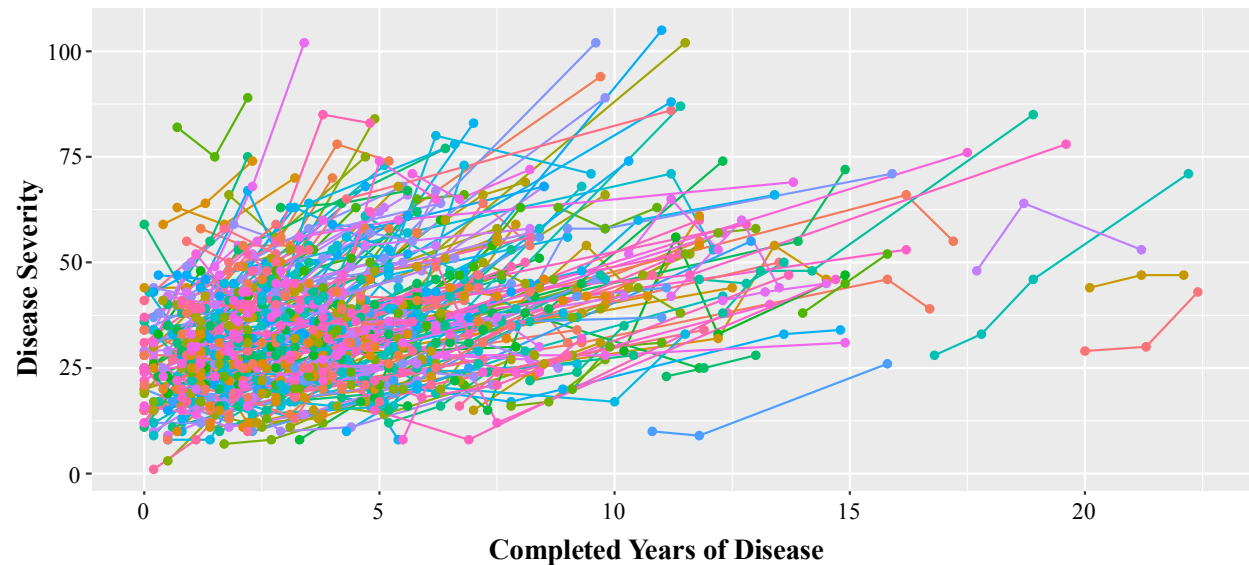
Parkinson's Disease (PD) is a **neurodegenerative** disorder characterized by immense **heterogeneity**. Due to the variation in disease severity and disease progression and a lack of understanding on accurate PD modeling, this study aimed at defining PD **subtypes** based on their disease progression and conducting a **predictive analysis** of these subtypes using baseline phenotypes. Due to the practical constraint of sample size, **extreme phenotypic analysis** was conducted to create the subtypes to boost the statistical power and simulate a larger sample size and **machine learning** (ML) approaches were utilized to conduct the predictive analysis. The study uncovered that subtypes could accurately create distinctions amongst a patient population and the ML model uncovered an accuracy of 78% when predicting the subtype from baseline characteristics. Furthermore, key clinical indicators were further validated such as age, Dyskinesias, and the overall severity index created from the Pairwise analysis as being the top endpoints from the ML model. This creates the possibility for **personalized clinical trials** where each PD subtype could receive different treatment and **multi-omics** data analysis using these subtypes could uncover biomarkers of PD disease progression.

Introduction

Parkinson's disease (PD) is a neurodegenerative motor disease of the nervous system (*Parkinson's Disease - Symptoms and Causes*, n.d.). Generally, PD and other neurodegenerative diseases such as Alzheimer's disease (AD) or frontotemporal dementias are coined as 'proteinopathies' due to their characteristic pathology of protein accumulation in the brain parenchyma (Bayer, 2013; Lam et al., 2024). Specifically, PD and other Lewy body disorders are 'synucleinopathic', characterized by the alpha-synuclein (aS)– aggregation in dopamine-producing neurons in the substantia nigra pars compacta (SNpc) which form clumps called Lewy bodies. Lewy bodies inhibit the production of dopamine which in turn leads to neuronal death (Srinivasan et al., 2021) creating the hallmark symptoms of PD primarily: motor symptoms like

tremor, bradykinesia (slow movement), postural instability and cognitive impairments of memory, language and visuospatial function (American Parkinson Disease Association, 2024).

Figure 1: Graph Displaying Heterogeneity of PD

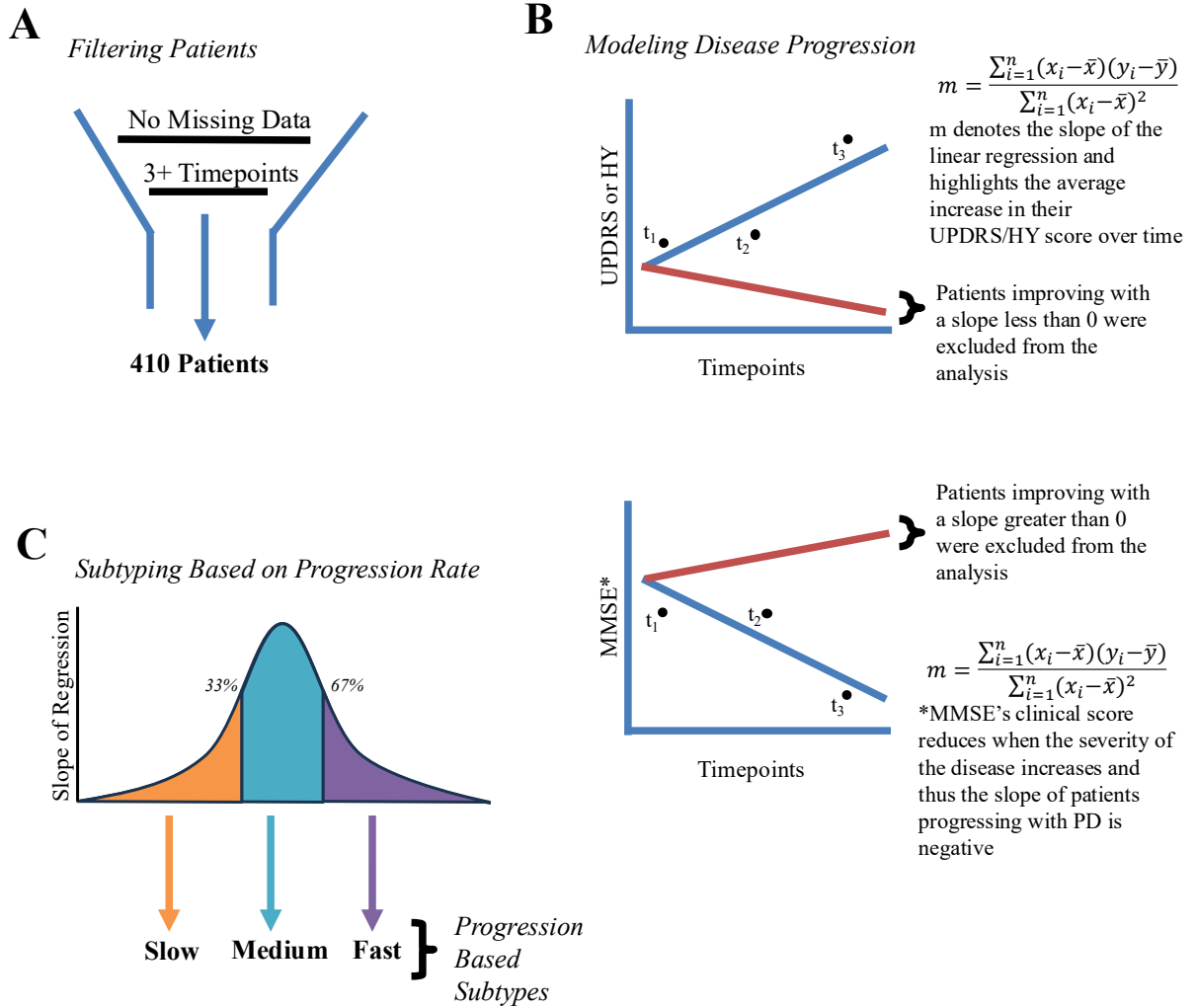


The wide range of symptoms coupled with its characteristic disease heterogeneity made it harder for physicians to diagnose PD and created a limited scope for preventive approaches and interventional treatment strategies. In other words, some patients may progress rapidly in terms of their motor symptom severity, while others have very little phenotypic variability where their symptoms remain constant for an extended period of time (see Figure 1). The reason for this variability is still unknown but is explained by the diverse aggregation of aS in different brain structures (Pagano et al., 2016). As a first step towards accelerating the discovery and validation of molecular and clinical markers for diagnosing and tracking early-stage PD, the Harvard Biomarker Study was initiated in 2007 (*HBS | AMP-PD*, n.d.). The Harvard Biomarker Study (HBS) is a longitudinal cohort of patients with neurodegenerative diseases such as PD or AD (n=16000) and collects bio samples such as cerebrospinal fluid (CSF), whole blood, plasma, and serum, becoming one of the largest biobanks for neurodegenerative diseases. This study sought to answer questions on PD heterogeneity, and how best disease progression can be predicted based on baseline characteristics.

Materials and Methods

Clinical Modeling of Patients and Creation of Subtypes

Figure 2: Schematic of overview of computational process of modeling and subtyping PD



All participants were patients (n=1600) recruited at the Brigham and Women's Hospital from 2007. Of this total, patients were filtered based on the quality and quantity of clinical data available. Patients that didn't have comprehensive data were excluded from the analysis (Figure 1A) and 410 patients were identified with significant data available. Specifically, patients with three or more timepoints of clinical data were included to boost statistical power and this ensured the reliability and accuracy of the patient stratified slopes (Cohen et al., 2011). There are numerous clinical scales used by physicians to assess the severity of PD for its wide-ranging

symptoms. This study utilized three clinical scales and separately used to assess progression of PD:

- Unified Parkinson's Disease Rating Scale (UPDRS), an additive scale combining 4 subparts assessing mainly motor and some non-motor symptoms on a scale of 1-108 (Unified Parkinson's Disease Rating Scale (UPDRS), n.d.).
- Hoehn and Yahr (HY), particularly assessing severity of motor symptoms on a scale of 1-7 including 1.5 and 2.5.
- Mini-Mental State Examination (MMSE), a cognitive screening used to assess overall cognitive function on a scale of 0-30 (Healthdirect Australia, n.d.).

$$m = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^n (x_i - \bar{x})^2}$$

When assessing the disease progression, a linear regression was fit on each of the three clinical scales assessed over their disease duration. Then the slope of this regression was extracted using the equation above where x_i and y_i are the individual data points and $\bar{x}$ and $\bar{y}$ are respectively the mean of the x-values and the mean of the y-values. This slope highlighted the change in the patient's symptomatic clinical severity over the time since their PD diagnosis. The equation above

Both UPDRS and HY are direct scales where the score increases corresponding to the severity of the symptoms whereas MMSE is an inverse scale and thus the score decreases as the severity of the symptoms increase. Patients showing improving symptoms were excluded as this study was targeted at subtyping patients progressing, not improving, with PD. Therefore, patients with negative slopes for UPDRS and HY and positive slope for MMSE were removed. Because different patients were either excluded or included depending on the clinical scale used, and each clinical scale assesses different forms of PD symptoms, three cohorts of progressors were created each for the clinical scale used to model the disease progression (Figure 1B).

Finally, patients were binned into subtypes based on their disease progression (informed by their slope of the regression). Each subtype is characterized by the tercile's of the slopes, such that the

highest slopes were fastest progressors, and the slopes in the lowest tercile were the slow progressors (Figure 1C). This subtyping is characteristic of extreme phenotypic analysis, where individuals with phenotypes of the extreme ends of a trait's distribution are selected. In this case, those individuals with extremely high and extremely low progression rates were identified. This allows for the study to find key clinical features that are predictive of the extreme phenotypes while boosting the statistical power of its assertions.

Study Population

Table 1: Overview of Demographics

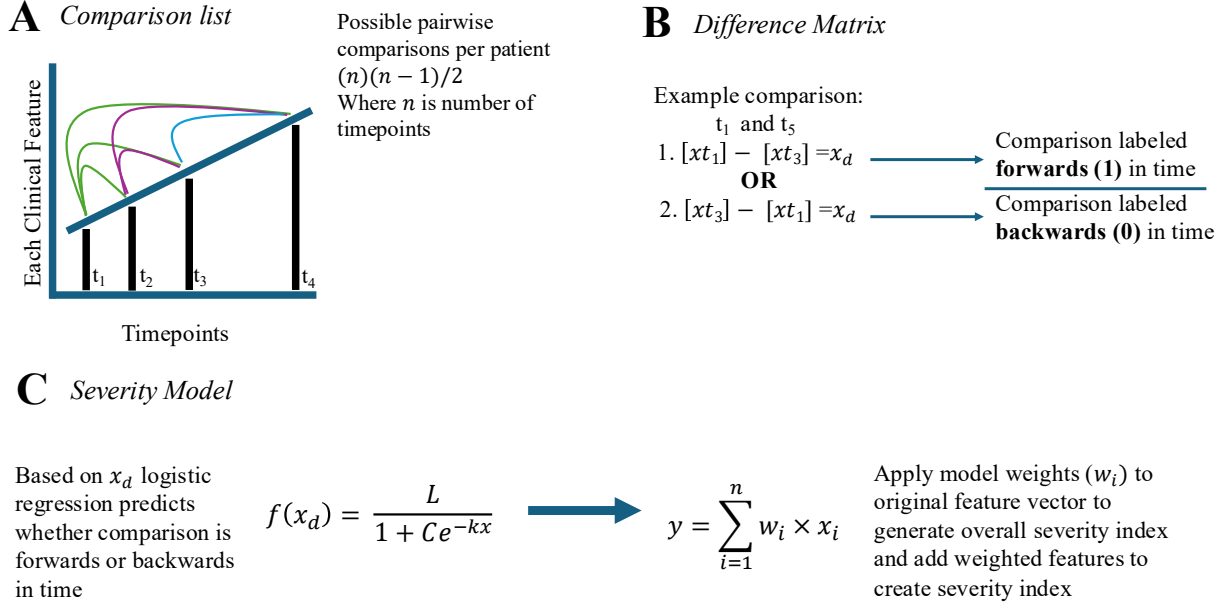
	Cohorts		
	UPDRS N = 327	Hoehn and Yahr N = 309	MMSE N = 242
Age at Baseline (years)*	65.7 ± 9.3	65.1 ± 10.0	65.6 ± 9
Sex (% female)	33.70%	35.70%	32.20%
Race (% white)	97.50%	97.40%	98.70%
Disease Duration at Baseline (years)*	2.85 ± 3.24	2.87 ± 3.25	2.70 ± 2.88

*Age and disease duration are presented as mean ± SD

The Institutional Review Boards approved the protocols for the recruitment and data usage which was performed with a written consent for participants.

Constructing Severity Index from Pairwise Analysis

Figure 3: Overview of Pairwise Analysis



The pairwise analysis using the method described by Gupta et al. (2023). Firstly, after the normalization of the dataset, all possible combinations between two timepoints for each individual patient were calculated. For each comparison, the difference between any clinical score is computed and assigned either assigned a forward's comparison ($t_1 - t_3$) or a backwards comparison ($t_3 - t_1$). Then, a logistic regression ML model was trained on this difference vector and tried to predict whether the comparison was forwards or backwards in time. The dataset was split into 70% training and a 30% holdout set. Ten-fold cross-validation approach was used to train and evaluate models with hyper tuning using L1 regularized penalty (λ). Accuracy metrics were calculated for each fold and the best-performing model was identified based on validation accuracy. Then, this model was evaluated on the holdout set to yield an accuracy. The Receiver Operating Characteristic (ROC) and the AUC (Area Under the Curve) were calculated to ensure sensitivity and specificity. Finally, feature importances were extracted by averaging the absolute coefficients of the logistic regression model across the folds. The top 5 feature weights were selected and multiplied in the original dataset, and for each timepoint they were added together to get an overall severity index per timepoint for each individual patient.

Predicting Based on Baseline Characteristics

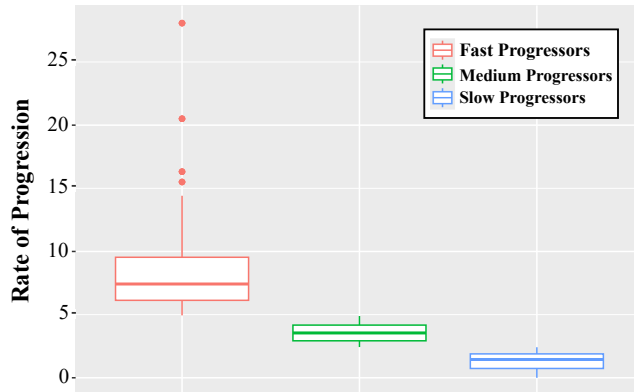
Finally, using the subtypes and the overall severity index from the pairwise, the baseline characteristics for the patients were extracted. Using all the physician assessments from each patient first visit, a classification model attempted to differentiate between the extreme subtypes (fast and slow progressors). The extreme phenotypes were used because it would offer the greatest statistical power with the minimal samples that the study had. Specifically, the data was split into two parts: 70% of the data was used in training and cross-validation, while the remaining 30% served as the holdout set. After hyperparameter tuning, a Random Forest Classifier model was used. Ten-fold cross-validation was applied and the best-performing fold determined by its testing accuracy was selected. This model was retrained on the full training dataset and subsequently evaluated on the holdout set. This approach was then utilized to predict the class of those patients progressing in terms of their UPDRS, MMSE and HY and thus the model was evaluated among all three cohorts.

Results

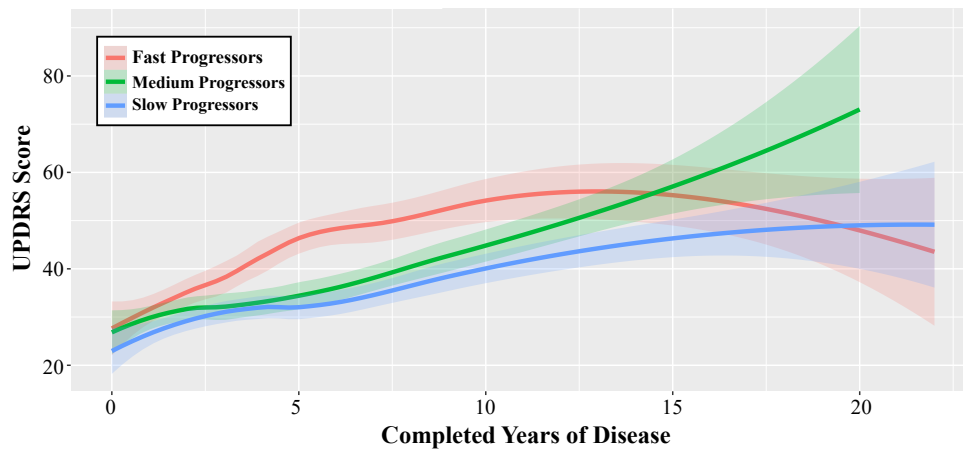
Modeling of Patient Subtypes

Figure 4: UPDRS Modeling and Subtyping

A Box and Whisker Plot of Subtypes vs. Rate of Progression of UPDRS Cohort



B Trend Line Plot of Completed Years of Disease vs. UPDRS Comparing Fast, Medium and Slow Subtypes



C Box and Whisker Plot of Completed Years of Disease vs. UPDRS Score Comparing Fast and Slow Subtypes

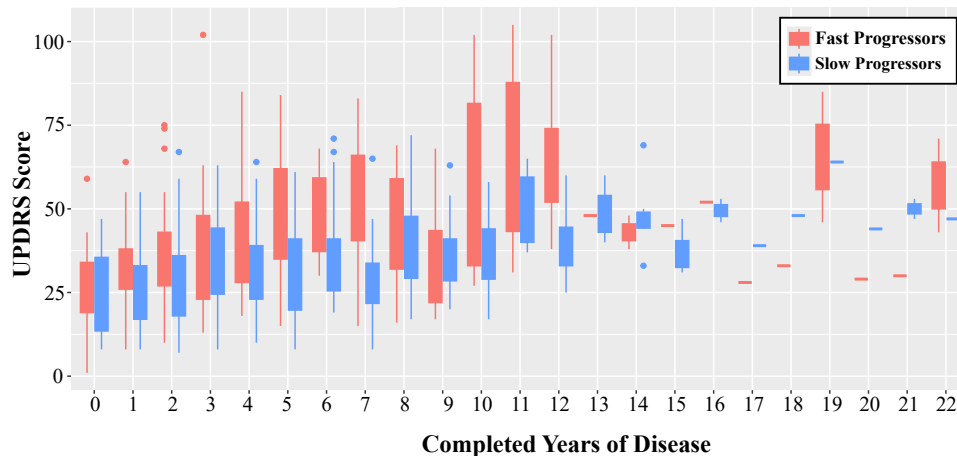
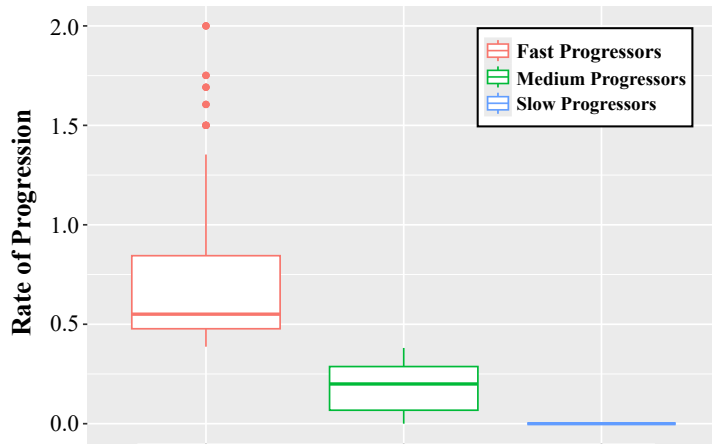


Figure 4A presents a box plot comparing the progression rate across the subtypes. This progression rate indicates the slope of the linear regression fit through the patients total UPDRS score and their completed years of disease and thus represents the average change in their UPDRS score per year. The ‘fast’ subtype exhibited the highest median slope ($M = 4.8$ UPDRS/ year, $IQR = 3.9-6.5$ UPDRS/ year), indicating greater increase in their motor symptoms over time compared to those patients binned in the ‘medium’ subtype ($M = 2.4$ UPDRS/ year, $IQR = 2.0-2.9$ UPDRS/ year) and ‘slow’ subtype ($M = 0.8$ UPDRS/ year, $IQR = 0.6-1.2$ UPDRS/ year). A significant difference was observed among the groups (Kruskal-Wallis H test, $p < 0.001$), with post-hoc analysis (Dunn’s test) revealing significant pairwise differences between all subtypes ($p < 0.001$) and their slope. Kruskal-Wallis H Test was used as the data wasn’t normal (Shapiro-Wilk Test $p < 0.05$). Outliers were noted in the fast subtype, where a patient had a slope of 19.3 UPDRS/ year significantly progressing over time.

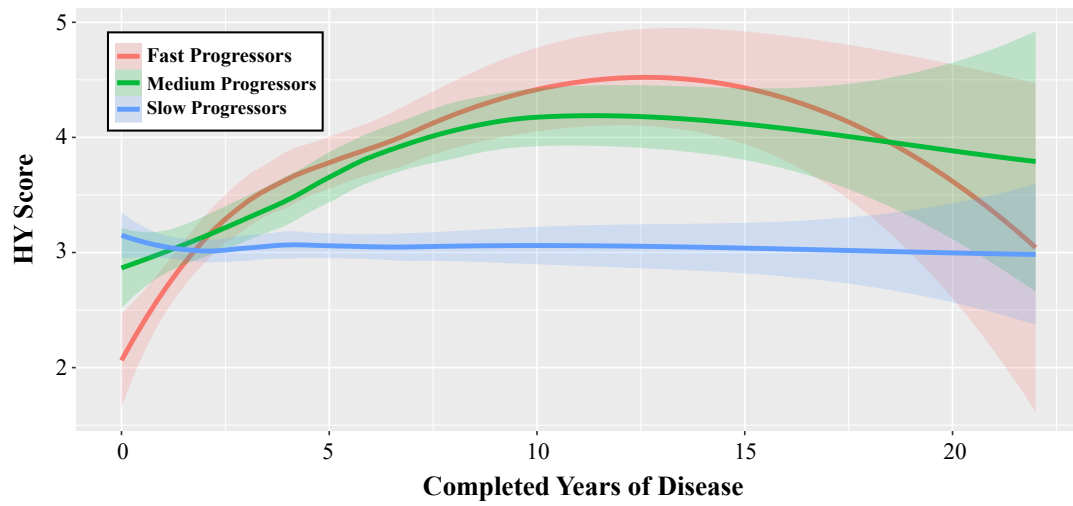
Furthermore, Figure 4B represents a trendline plot of the UPDRS scores of each subtype. This plot fits a smooth curve on each of the subtypes and the shaded region represents 95% confidence intervals, indicating the uncertainty in the estimated trends. Figure 4C presents a series of boxplots representing the UPDRS scores of the extreme subtypes (fast and slow progressors). At baseline it was noted that fast progressors start with a higher UPDRS score than slow progressors, but that pattern shifts after 18 years from diagnosis. Post hoc analysis using Dunn’s test confirmed that there were significant pairwise differences among the extreme subtypes ($p < 0.001$) whereas all three phenotypes differentiate less significantly ($p = 0.03$). When the completed year of disease exceeds 12, the number of patients per timepoint significantly drops ($n \sim 6$) and later timepoints revealed only one patient per timepoint, thus produces unreliable data. Hence, there are no significant differences between any of the three subtypes ($p > 0.05$) beyond 14 years.

Figure 5: HY Modeling and Subtyping

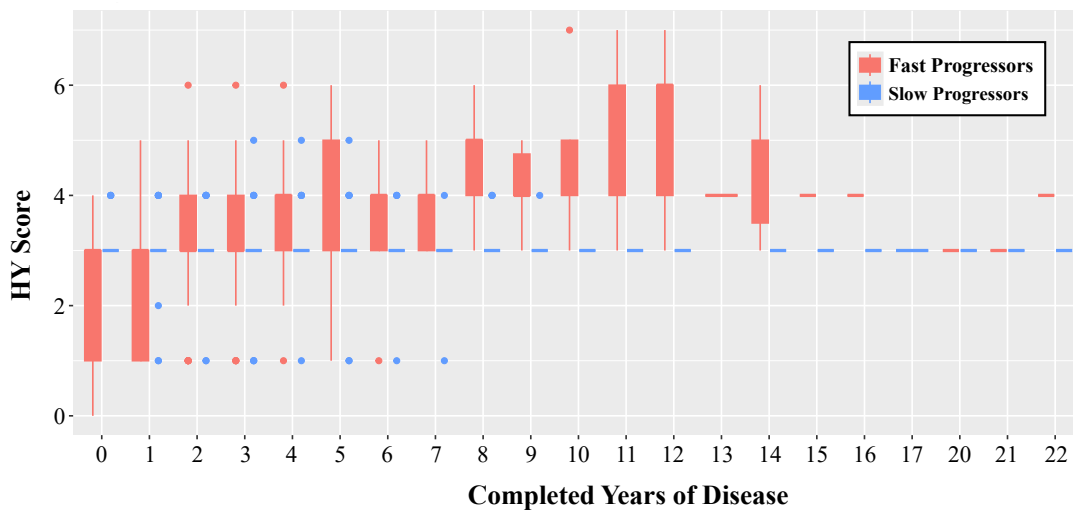
A Box and Whisker Plot of Subtypes vs. Rate of Progression of HY Cohort



B Trend Line Plot of Completed Years of Disease vs. HY Comparing Fast, Medium and Slow Subtypes



C Box and Whisker Plot of Completed Years of Disease vs. HY Score Comparing Fast and Slow Subtypes

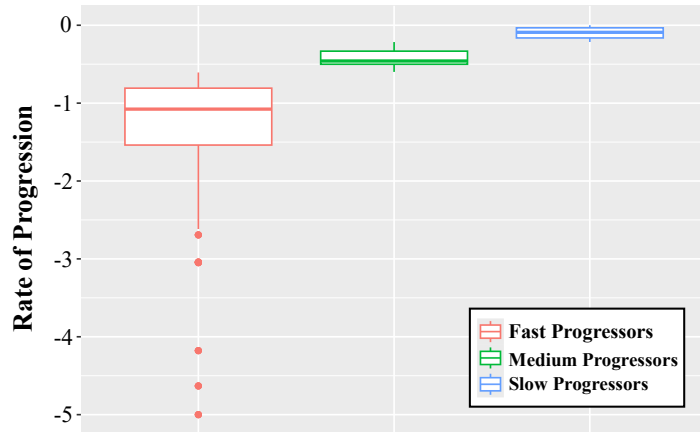


Furthermore, in a similar analysis for those patients progressing in terms of their HY score, Figure 5A presents a box plot comparing the progression rate across the subtypes. The ‘fast’ subtype exhibited the highest median slope ($M = 0.36$ HY/ year, $IQR = 0.29-0.55$ HY/ year), indicating greater increase in their motor symptoms over time compared to those patients binned in the ‘medium’ subtype ($M = 0.13$ HY/ year, $IQR = 0.07-0.19$ HY/ year) and ‘slow’ subtype ($M = 0$, $IQR = 0$). A significant difference was observed among the groups (Kruskal-Wallis H Test, $p < 0.001$), with Dunn’s test revealing significant pairwise differences between all subtypes ($p < 0.001$) and their slope. Kruskal-Wallis H Test was used as the data wasn’t normal (Shapiro-Wilk Test $p < 0.05$). Outliers were noted in the fast subtype, with multiple patients that had slopes above 1.5 HY/ year significantly progressing.

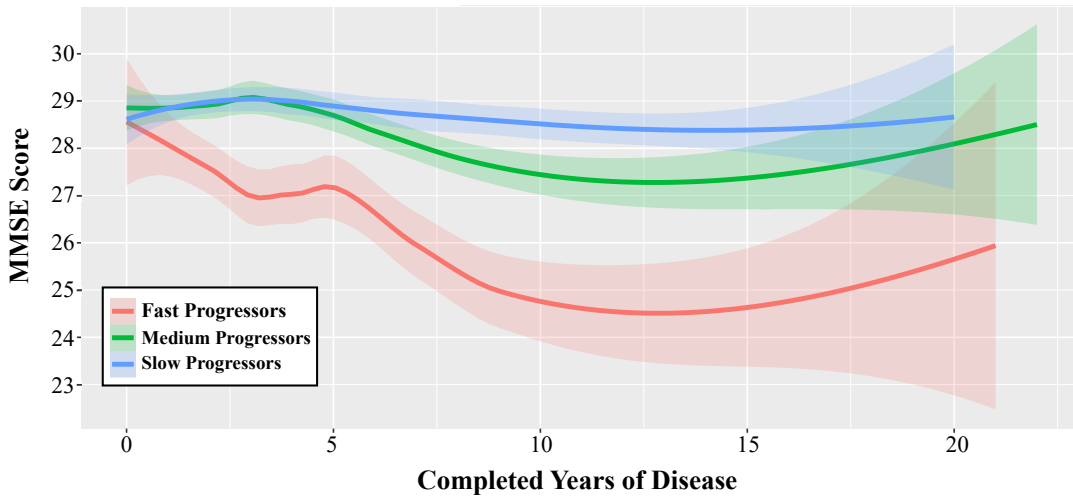
Furthermore, Figure 5B represents a trendline plot of HY scores of each subtype. This plot fits a smooth curve on each of the subtypes and the shaded region represents 95% confidence intervals, indicating the uncertainty in the estimated trends. Figure 5C presents a series of boxplots representing the HY scores of the extreme subtypes (fast and slow progressors). At baseline it was noted that fast progressors start with a lower HY score than slow progressors, but that pattern shifts after 18 years from diagnosis. Post hoc analysis using Dunn’s test confirmed that there were significant pairwise differences among the extreme subtypes ($p < 0.001$) whereas all three phenotypes differentiate less significantly ($p = 0.04$). Outliers were observed initially for fast progressors at years 0 and 1 due to the natural growth of HY over time. Similar to UPDRS, after the completed years of disease exceeds 12, the number of patients per timepoint significantly drops ($n \sim 5$) which thus produces unreliable data.

Figure 6: MMSE Modeling and Subtyping

A Box and Whisker Plot of Subtypes vs. Rate of Progression of MMSE Cohort



B Trend Line Plot of Completed Years of Disease vs. UPDRS Comparing Fast, Medium and Slow Subtypes



C Box and Whisker Plot of Completed Years of Disease vs. UPDRS Score Comparing Fast and Slow Subtypes

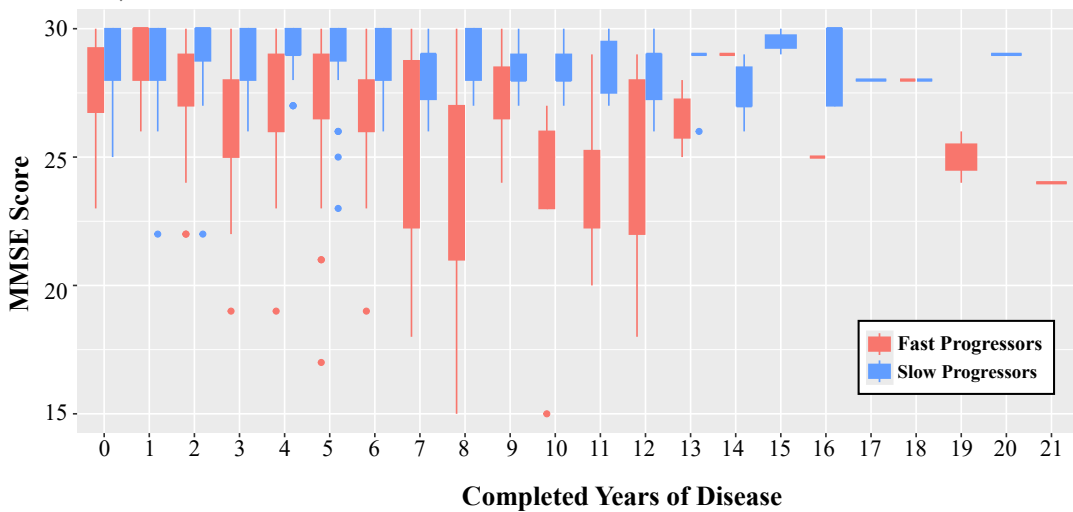


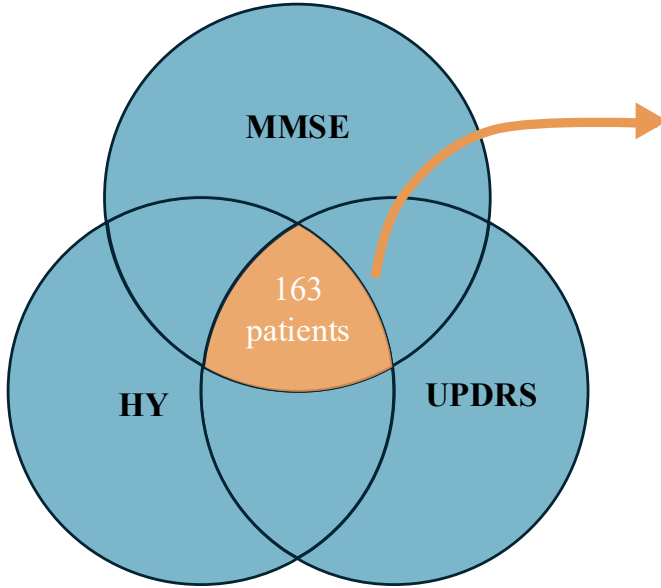
Figure 6A similarly presents a box plot comparing the progression rate across the subtypes. The ‘fast’ subtype exhibited the highest median score ($M = -0.74$ MMSE / year, $IQR = [-0.55) - (-1.05)$ MMSE / year], indicating greater increase in their cognitive symptoms over time compared to those patients binned in the ‘medium’ subtype ($M = -0.27$ MMSE / year, $IQR = [-0.22) - (-0.35)$ MMSE / year] and ‘slow’ subtype ($M = -0.07$ MMSE / year, $IQR = [(-0.03) - (-0.11)$ MMSE / year]. A significant difference was observed among the groups ($p < 0.001$), with Dunn’s test revealing significant pairwise differences between all subtypes ($p < 0.001$) and their slope. Kruskal-Wallis H Test was used as the data wasn’t normal (Shapiro-Wilk Test $p < 0.05$). Outliers were noted in the fast subtype, with multiple patients that had slopes from (-2) MMSE/ years to (-3) MMSE/ years which were patients that are significantly progressing. As stated in the *methods and materials* section, MMSE slopes are all negative as patients with a lower MMSE score performed cognitively worse than higher scores.

Furthermore, Figure 6B represents a trendline plot of MMSE scores of each subtype. This plot fits a smooth curve on each of the subtypes and the shaded region represents 95% confidence intervals, indicating the uncertainty in the estimated trends. Figure 6C presents a series of boxplots representing the HY scores of the extreme subtypes (fast and slow progressors). At baseline, no significant differences were observed between the extreme phenotypes, however as time progressed, they began to differentiate more significantly. Furthermore, post hoc analysis using Dunn’s test confirmed that there were significant pairwise differences among the extreme subtypes ($p < 0.001$) whereas all three phenotypes differentiate less significantly ($p = 0.01$). Outliers were observed after the completed years of disease exceeds 12, the number of patients per timepoint significantly drops ($n \sim 6$) which thus produces unreliable data.

Understanding The Clinical Scales: UPDRS, HY and MMSE

Figure 7: Creation of extreme progressors and correlation between clinical scales

A Severe Patient Progressors Identified



B Features Identified

Rank	Feature
1	Age
2	Disease Duration
3	Toe Tapping
4	Dressing
5	Falling
6	MMSE Total Score
7	Gender
8	Dyskinesias
9	Subtotal 2
10	Posture

T tests were conducted to determine the statistically significant features that differentiate severe progressors from normal ones

C Correlation Between Three Scales

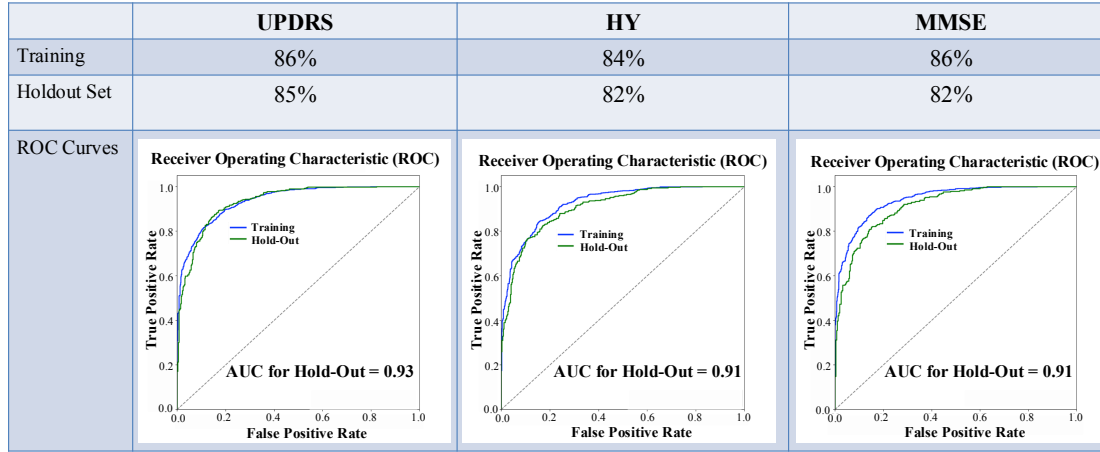
	UPDRS	MMSE	HY
UPDRS	1.00	-0.31	0.61
MMSE	-0.31	1.00	-0.24
HY	0.61	-0.24	1.00

As highlighted by figures 4, 5 and 6 by the variation in trends in each clinical scale, disease severity can't be described by a single clinical scale. In an effort to understand the relationship between these scales, severe progressors were initially identified. 163 patients had all three clinical scales worsening over time and were identified as the severe phenotype. To understand the clinical features that differentiate these severe progressors from patients only progressing in terms of one or two clinical scales, T- tests were conducted and the top ten features were identified with its top being age and disease duration. Furthermore, the correlation between all three scales was calculated to understand the relationship between the scales, where UPDRS and HY correlated the highest at 0.61 followed by MMSE and UPDRS at -0.31 and HY and MMSE with a correlation of -0.24.

Constructing Severity Index from Pairwise Analysis

Figure 8: Pairwise Analysis Results

A Testing and Training Accuracy



B Top 5 Feature Weights Per Cohort

UPDRS		HY		MMSE	
Endpoint	Weight	Endpoint	Weight	Endpoint	Weight
Dyskinesias	3.29	Final HY score	6.35	Dyskinesias	4.35
Dyskinesias	2.86	Memory	3.18	Counting	3.35
Freezing When Walking	2.72	Dyskinesias	3.09	Date Recall	2.91
Dyskinesias	2.47	Thought Disorder	2.87	Speech	2.87
Posture	2.37	Posture	2.65	Final MMSE Score	2.86

The logistic regression's model performance was evaluated across the three cohorts of progressors, UPDRS, MMSE and HY. In Figure 6A, training accuracies were the highest for the UPDRS cohort (86%) and MMSE (86%) compared to 84% for the HY cohort. Hold-out accuracies were slightly lower than the training accuracies where UPDRS had the highest accuracy (85%) followed by HY and MMSE (both 82%). ROC curves were generated for each cohort with AUC values which reveals the discriminatory power of distinguishing between the positive and negative classes (values of 0.5 indicates the model performs no better than guessing). The ROC curves of both training and testing for all cohorts present that the model didn't overfit because there was no significant drop in the hold-out set's accuracy. The UPDRS cohort achieved the highest (AUC = 0.93) followed by HY and MMSE (AUC = 0.91).

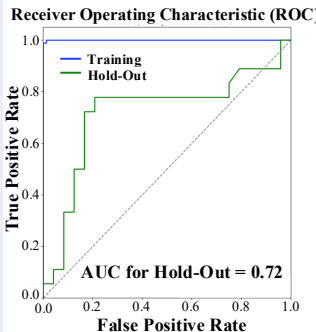
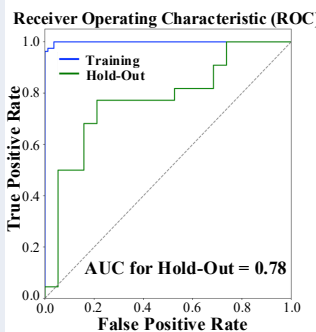
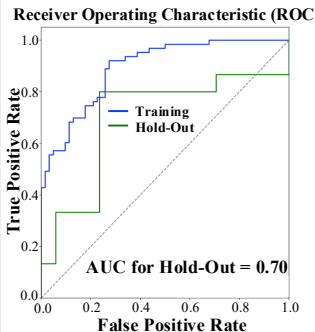
Feature importance was extracted from the model highlighting the most influential predictors across each cohort. Figure 6B highlights the top 5 predictors for each cohort, where Dyskinesias

consistently was in the top three for each cohort. The feature weights were then used to assemble a severity index for each patient and was tested to differentiate in the ML model at baseline.

Accurate Classification of Subtypes Based on Baseline Features

Figure 9: Baseline Classification Results

A Extreme Subtype ML Classification

	Fast vs Slow Prediction											
	UPDRS			HY			MMSE					
Holdout Set	78.57%			78.05%			78.12%					
Confusion Matrix	Predicted			Predicted			Predicted					
		Fast	Slow		Fast	Slow		Fast	Slow			
	Actual	Fast	19	5	Actual	Fast	15	3	Actual	Fast	13	4
		Slow	4	14		Slow	5	17		Slow	3	12
ROC Curves	Receiver Operating Characteristic (ROC)			Receiver Operating Characteristic (ROC)			Receiver Operating Characteristic (ROC)					
												

B Top 5 Feature Weights Per Cohort

UPDRS		HY		MMSE	
Endpoint	Weight	Endpoint	Weight	Endpoint	Weight
Age	0.07	Age	0.07	Age	0.32
Total UPDRS	0.06	Signed Severity Index*	0.06	Bradykinesia	0.21
Subscale 4	0.06	Hoehn and Yahr	0.06	Subscale 3	0.07
Disease Duration	0.05	Absolute Severity Index	0.05	Bradykinesia	0.04
Subscale 3	0.04	Disease Duration	0.04	Handwriting	0.03

*The summative severity index from pairwise analysis included positive and negative weights, while the absolute severity index used their absolute values, and the signed severity index used both positive and negative weights.

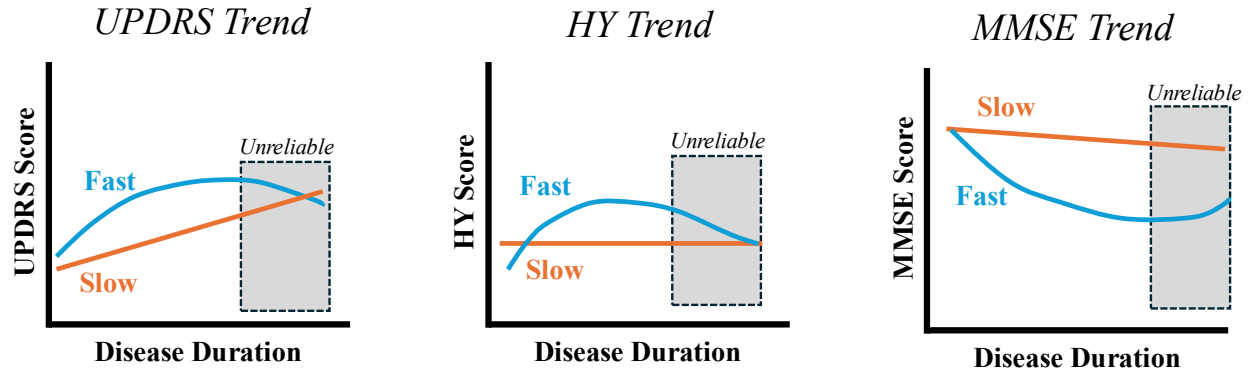
Coinciding with extreme phenotypic analysis, the model was tasked with distinguishing between the extreme phenotypes (fast and slow progressors). The model achieved hold-out accuracies of

78% across UPDRS, HY and MMSE patient progressors (Figure 9A). Figure 9A also details the confusion matrices for each cohort which demonstrate the specific number of false positives and true negatives that the model identified. To further evaluate the model's discriminatory power, ROC curves were generated for each cohort. HY cohort achieved the highest AUC of 0.78 followed by UPDRS (0.72) and MMSE (0.70). Figure 7B reveals the individual feature weights assigned by the model for each cohort. Even though the cohorts reflected a unique clinical emphasis of each scale, across all three cohorts, age was considered the most important feature. Also, UPDRS subscale 3 (which is a summative subscale of motor symptoms), the disease duration of the patient and the severity index's (both signed and absolute) were among the top features identified by the model.

Discussion

This study identified key findings, including the stratification of patient progression rates based on clinical scale endpoints that enables the classification of patients into progression subtypes (fast, medium, and slow). Considering that disease severity can't be described by a single clinical scale, this study clarified the relationship between three key clinical scales (UPDRS, MMSE and HY). This investigation also allowed discovery of influential subscale features predictive of disease severity, which were used to develop a novel severity index representing a significant step toward simplifying and enhancing clinical assessment in PD. Finally, a predictive analysis was performed, attempting to differentiate between the extreme phenotypes (fast and slow progressors) based on baseline characteristics where age was identified as the most predictive factor followed by the severity index identified from the pairwise analysis.

Figure 10: Summary of Trends for Each Clinical Scale



We observed that the subtyping of PD based on the progression rate can be done accurately, and that extreme subtypes (fast and slow progressors) differentiate more significantly than the intermediate medium progressors (Figures 3-5). It was also noted that all three scales showed a different trend of these subtypes as time progressed summarized by Figure 10. Fast progressors in the HY cohort presented a lower HY score with less severe motor symptoms at baseline compared to slow progressors. Conversely, patients binned in the fast subtype in the MMSE cohort had worse cognitive symptoms at baseline than slower progressors. This finding suggests a critical pattern where rapid motor progression may initially align with milder motor symptoms but worse cognitive impairments. Furthermore, clinical data beyond 12–13 years of disease duration introduced significant noise, potentially due to limited sample sizes at these later timepoints and thus trend analysis at these timepoints was unreliable.

When analyzing the relationship between these three clinical scales, it was revealed that UPDRS and HY correlate positively (0.61) as both assess primarily motor symptoms whereas when correlated with MMSE, this correlation drops. Also, age and disease duration were identified as the greatest differentiators from severe patient progressors (progressing in all three clinical scales) compared with milder progressors (progressing in one or two scales).

The pairwise analysis identified key clinical features contributing to disease severity across UPDRS, MMSE, and HY cohorts. Unlike HY which is single – scale score, UPDRS and MMSE are both summative scales, comprising of numerous clinical features (e.g., tremor, bradykinesia, rigidity, etc.) that were assessed by the physician and totaled to give the final score. UPDRS assesses 60 clinical features and MMSE assesses 20 clinical features which are time-consuming

to quantify by physicians and may be unnecessary in assessing general PD severity. The pairwise analysis revealed that dyskinesias is one the most influential clinical features when assessing severity of symptoms as well as their HY score, and some cognitive endpoints like date recalling and counting. Using the top five feature weights, a streamlined severity index was developed, offering a practical alternative to the cumbersome UPDRS and MMSE scales for assessing patient severity.

Finally, the predictive model was able to accurately use the results from the previous analyses involving the subtypes from the clinical modeling and the severity index from the pairwise analysis. The accurate differentiation of the extreme subtypes based on characteristics assessed on the first visit of all patients highlights the true potential for personalized clinical trial enrolment and target-based analysis to uncover the biological mechanism behind the extreme variation in progression rates of PD. Also, the severity index from the Pairwise analysis was further validated by the model as being one of the top differentiators among the classes. This highlights the potential for the severity index calculated to act as a clinical marker for the progression-based subtypes and could facilitate personalized treatment approaches based on their baseline disease and their corresponding progression rate. Also, this highlights the potential for further multi-omics screening (proteomics / genomic) to reveal biomarkers that are indicative of PD subtypes. Also, this predictive model highlighted age as the top feature for the baseline predictions, which offers the possible explanation for the heterogeneity of progression rates. This signals that future analyses may want to correct for age to further accurately create patient stratified slopes to model progression.

However, the consistent accuracy, particularly in the HY cohort, suggests that patients progressing in terms of their HY score are more accurately binned into their according subtype than UPDRS or MMSE. This finding goes against the typical notion, as HY score is considered a crude physician assessed scale, and previous studies have identified that UPDRS is the newest and best scale to assess PD (Perlmutter, 2009).

While many studies such as Fereshtehnejad et al. (2017) or Vavougiou et al. (2018) model PD disease subtypes, they analyze data from the Parkinson's Progression Markers Initiative (PPMI).

PPMI typically enrolls patients with a restrictive selection criterion (e.g., enrolling early stage, untreated PD patients) which limits generalizability to a diverse population (Study Cohorts, n.d.). This study used clinical data from the Harvard Biomarker Study, which includes a diverse patient population, and thus offers greater applicability. Furthermore, no previous studies have used the pairwise analysis to rank the UPDRS and MMSE subscale features based on severity and used the feature weights to create a severity index for PD. However, in a study by Oğul and Özdemir (2022), they use a pairwise ranking methodology, to specifically rank gait-based symptoms. While modeling and subtyping of PD based on progression rates has been done in prior studies, this study differentiates between the three key measures of progression (UPDRS, HY and MMSE) and identifies their inter-relationships when assessing symptoms.

However, this study had some limitations. Firstly, the sample size for each cohort was relatively small. From table 1, UPDRS and HY cohort respectively had 318 and 309 patients, and MMSE had 236. Furthermore, from table 1, the overwhelming majority of participants were white which thus reduces clinical generalizability to other demographic groups. Lastly, the features identified by the pairwise analysis and the accurate subtyping of PD based on their progression rate should be validated in another multi centered longitudinal cohort of PD patients. Future research should focus on validating these findings across diverse cohorts and integrating genetic and proteomic data to identify biomarkers indicative of progression rates.

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